UNIT - IV

INVERSE LAPLACE TRANSFORM

Now we obtain $f(t)$ when $\phi(s)$ is given, then we say that inverse Laplace transform of $\phi(s)$ is $f(t)$.

(1) If $L[f(t)] = \phi(s)$, then $L^{-1}[\phi(s)] = f(t)$

where L^{-1} is called the inverse Laplace transform operator.

(2) If $\varphi_1(s)$ and $\varphi_2(s)$ are L.T. of $f(t)$ and $g(t)$ respectively then

$$L^{-1}[C_1\varphi_1(s) + C_2\varphi_2(s)] = C_1L^{-1}[\varphi_1(s)] + C_2L^{-1}[\varphi_2(s)]$$

Proof : Given : $L[f(t)] = \varphi_1(s)$

$$f(t) = L^{-1}[\varphi_1(s)]$$

$$L[g(t)] = \varphi_2(s)$$

$$g(t) = L^{-1}[\varphi_2(s)]$$

W.K.T. $L[C_1f(t) + C_2g(t)] = C_1L[f(t)] + C_2L[g(t)]$

$$= C_1\varphi_1(s) + C_2\varphi_2(s)$$

$$C_1f(t) + C_2g(t) = L^{-1}[C_1\varphi_1(s) + C_2\varphi_2(s)]$$

i.e., $L^{-1}[C_1\varphi_1(s) + C_2\varphi_2(s)] = C_1f(t) + C_2g(t)$

$$= C_1L^{-1}\varphi(s) + C_2L^{-1}\varphi_2(s)$$

Note : (1) If $L[f(t)] = \varphi(s)$ then $L[e^{at}f(t)] = \varphi(s-a)$

i.e., If $L^{-1}[\varphi(s)] = f(t)$ then

$$L^{-1}[\varphi(s-a)] = e^{at}f(t) = e^{at}L^{-1}[\varphi(s)]$$

Note : (2) If $L[f(t)] = \varphi(s)$ then $L[e^{-at}f(t)] = \varphi(s+a)$

i.e., If $L^{-1}[\varphi(s)] = f(t)$ then

$$L^{-1}[\varphi(s+a)] = e^{-at}f(t) = e^{-at}[L^{-1}\varphi(s)]$$

♦ IMPORTANCE FORMULA

1. $L^{-1}\left[\frac{1}{s}\right] = 1$

2. $L^{-1}\left[\frac{1}{s^n}\right] = \frac{t^{n-1}}{\lfloor n-1}$

3. $L^{-1}\left[\frac{1}{s-a}\right] = e^{at}$

4. $L^{-1}\left[\frac{s}{s^2-a^2}\right] = \cosh at$

5. $L^{-1}\left[\frac{1}{s^2-a^2}\right] = \frac{1}{a}\sinh at$

6. $L^{-1}\left[\frac{1}{s^2+a^2}\right] = \frac{1}{a}\sin at$

7. $L^{-1}\left[\frac{s}{s^2+a^2}\right] = \cos at$

8. $L^{-1}\left[F(s-a)\right] = e^{at} f(t)$

9. $L^{-1}\left[\frac{1}{(s-a)^2+b^2}\right] = \frac{1}{b} e^{at}\sin b\,t$

10. $L^{-1}\left[\frac{s-a}{(s-a)^2+b^2}\right] = e^{at}\cos bt$

11. $L^{-1}\left[\frac{1}{(s-a)^2-b^2}\right] = \frac{1}{b} e^{at}\sinh bt$

12. $L^{-1}\left[\frac{s-a}{(s-a)^2-b^2}\right] = e^{at}\cosh bt$

13. $L^{-1}\left[\frac{1}{(s^2+a^2)^2}\right] = \frac{1}{2a^3}(\sin at - at\cos at)$

14. $L^{-1}\left[\frac{s}{(s^2+a^2)^2}\right] = \frac{1}{2a} t\sin at$

15. $L^{-1}\left[\frac{s^2-a^2}{(s^2+a^2)^2}\right] = t\cos at$

16. $L^{-1}[1] = \delta(t)$

17. $L^{-1}\left[\frac{s^2}{(s^2+a^2)^2}\right] = \frac{1}{2a}[\sin at + at\cos at]$

Example 1 Find $L^{-1}\left[\dfrac{2s+1}{s^2+4s+13}\right]$

Solution : $L^{-1}\left[\dfrac{2s+1}{s^2+4s+13}\right] = L^{-1}\left[\dfrac{2s+1}{(s+2)^2+3^2}\right]$

$$= L^{-1}\left[\frac{2s+4-3}{(s+2)^2+3^2}\right]$$

$$= L^{-1}\left[\frac{2(s+2)}{(s+2)^2+3^2}\right] - L^{-1}\left[\frac{3}{(s+2)^2+3^2}\right]$$

$$= 2e^{-2t}L^{-1}\left[\frac{s}{s^2+3^2}\right] - e^{-2t}L^{-1}\left[\frac{3}{s^2+3^2}\right]$$

$$= 2e^{-2t}\cos 3t - e^{-2t}\sin 3t$$

Example 2 Find $L^{-1}\left[\dfrac{1}{(s^2+a^2)(s^2+b^2)}\right]$

Solution : $L^{-1}\left[\dfrac{1}{(s^2+a^2)(s^2+b^2)}\right]$

$$= \frac{1}{b^2-a^2}L^{-1}\left[\frac{b^2-a^2}{(s^2+a^2)(s^2+b^2)}\right]$$

$$= \frac{1}{b^2-a^2}L^{-1}\left[\frac{1}{s^2+a^2} - \frac{1}{s^2+b^2}\right]$$

$$= \frac{1}{b^2-a^2}\frac{1}{a}L^{-1}\left[\frac{a}{s^2+a^2}\right] - \frac{1}{b^2-a^2}\frac{1}{b}L^{-1}\left[\frac{b}{s^2+b^2}\right]$$

$$= \frac{1}{a(b^2-a^2)}\sin at - \frac{1}{b(b^2-a^2)}\sin bt$$

$$= \frac{1}{b^2-a^2}\left[\frac{1}{a}\sin at - \frac{1}{b}\sin bt\right]$$

♦ MULTIPLICATION BY s

If $\quad L[f(t)] = \varphi(s)$ then

$$L[f'(t)] = SL[f(t)] - f(0)$$

i.e., If $L^{-1}[\varphi(s)] = f(t)$

then $L^{-1}[s\varphi(s)] = f'(t)$

$$= \frac{d}{dt}f(t) = \frac{d}{dt}[L^{-1}\varphi(s)]$$

Provided $f(0) = 0$, $L^{-1}[f(0)] = 0$

when $t \to 0$

$$L^{-1}[s F(s)] = \frac{d}{dt}f(t) + f(0)\,\delta(t)$$

PROBLEMS BASED ON INVERSE L.T. MULTIPLICATION BY s

Example 1 **Find** $L^{-1}\left[\dfrac{s}{s^2+1}\right]$

Solution : $L^{-1}\left[\dfrac{1}{s^2+1}\right] = \sin t$

$$L^{-1}\left[\frac{s}{s^2+1}\right] = \frac{d}{dt}(\sin t) + \sin(0)\,\delta(t) = \cos t$$

Example 2 **Find** $L^{-1}\left[\dfrac{s}{4s^2-25}\right]$

Solution : $L^{-1}\left[\dfrac{1}{4s^2-25}\right] = \dfrac{1}{4}L^{-1}\left[\dfrac{1}{s^2-\dfrac{25}{4}}\right]$

$$= \frac{1}{4}\,\frac{2}{5}L^{-1}\left[\frac{5/2}{s^2-(5/2)^2}\right] = \frac{1}{10}\sinh\frac{5}{2}t$$

$$L^{-1}\left[\frac{s}{4s^2-2t}\right] = \frac{d}{dt}\left[\frac{1}{10}\sinh\frac{5}{2}t\right] + \frac{1}{10}\sinh\frac{5}{2}(0)f(t)$$

$$= \frac{1}{10}\cosh t\frac{5}{2}\left(\frac{5}{2}\right) + 0 = \frac{1}{4}\cosh\frac{5}{2}t$$

♦ **MULTIPLICATION BY 1/S.**

PROBLEMS BASED ON INVERSE L.T. [Multiplication by 1/s]

Example 3 If $L[f(t)] = \varphi(s)$, then $L\left[\int_0^t f(t)\,dt\right] = \frac{1}{s}\varphi(s)$

i.e., $L^{-1}\left[\frac{1}{s}\varphi(s)\right] = \int_0^t f(t)\,dt = \int_0^t L^{-1}[\varphi(s)]\,dt$

Proof : $L[f(t)] = \int_0^\infty e^{-st} f(t)\,dt$

$$L\left[\int_0^t f(t)\,dt\right] = \int_0^\infty e^{-st}\left[\int_0^t f(t)\,dt\right]dt$$

$$= \int_0^\infty \left[\int_0^t f(t)\,dt\right] d\left[\frac{e^{-st}}{-s}\right]$$

$$= \left[\left(\int_0^t f(t)\,dt\right)\left(\frac{e^{-st}}{-s}\right)\right]_0^\infty - \int_0^\infty \left(\frac{e^{-st}}{-s}\right) f(t)\,dt$$

[$\because$ diff $\int f(t)\,dt = f(t)$]

$$= (0-0) + \frac{1}{s}\int_0^\infty e^{-st} f(t)\,dt$$

$$= \frac{1}{s}L[f(t)] = \frac{1}{s}\varphi(s)$$

Example 4 **Find** $L^{-1}\left[\frac{1}{s(s+3)}\right]$

Solution : $L^{-1}\left[\frac{1}{s(s+3)}\right] = \int_0^t L^{-1}\left[\frac{1}{s+3}\right]dt$

$$= \int_0^t e^{-3t}\,dt = \left[\frac{e^{-3t}}{-3}\right]_0^t = -\frac{1}{3}\left[e^{-3t}\right]_0^t$$

$$= -\frac{1}{3}\left[e^{-3t}-1\right] = \frac{1}{3}\left[1-e^{-3t}\right]$$

♦ INVERSE LAPLACE TRANSFORMS OF DERIVATIVES.

W.K.T. If $L[f(t)] = \varphi(s)$ then $L[t f(t)] = -\varphi'(s)$

i.e., If $L^{-1}[\varphi(s)] = f(t)$ then $L^{-1}[\varphi'(s)] = -t f(t)$

$$= -t\, L^{-1}[\varphi(s)]$$

Example 5. **Find** $L^{-1}\left[\frac{s}{(s^2-a^2)^2}\right]$

Solution : Let $\varphi'(s) = \frac{s}{(s^2-a^2)^2}$

$$\begin{array}{|l|}\hline \text{Put } s^2 - a^2 = t \\ 2s\,ds = dt \\ s\,ds = \frac{dt}{2} \\ \hline\end{array}$$

$$\int \varphi'(s)\,ds = \int \frac{s}{(s^2-a^2)^2}\,ds$$

$$\varphi(s) = \int \frac{s}{(s^2-a^2)^2}\,ds$$

$$\therefore \varphi(s) = \int \frac{1}{t^2}\frac{dt}{2} = \frac{1}{2}\left[\frac{-1}{t}\right] = -\frac{1}{2t} = -\frac{1}{2(s^2-a^2)}$$

W.K.T. $L^{-1}(\varphi'(s)] = -t\, L^{-1}[\varphi(s)]$

$$= -t\, L^{-1}\left[\frac{-1}{2(s^2-a^2)}\right] = \frac{t}{2} L^{-1}\left[\frac{1}{(s^2-a^2)}\right]$$

$$= \frac{t}{2}\frac{1}{a} L^{-1}\left[\frac{a}{s^2-a^2}\right] = \frac{t}{2a}\sinh at$$

Example 6 **Find** $L^{-1}\left[\tan^{-1}\left(\frac{1}{s}\right)\right]$

Solution : W.K.T. $L^{-1}[\varphi'(s)] = -t\, L^{-1}[\varphi(s)]$

$$L^{-1}[\varphi(s)] = -\frac{1}{t} L^{-1}[\varphi'(s)]$$

$$L^{-1}\left[\tan^{-1}\left(\frac{1}{s}\right)\right] = -\frac{1}{t} L^{-1}\left[\frac{d}{ds}\tan^{-1}\left(\frac{1}{s}\right)\right]$$

$$= -\frac{1}{t} L^{-1}\left[\frac{1}{1+\frac{1}{s^2}}\left(\frac{-1}{s^2}\right)\right]$$

$$= -\frac{1}{t} L^{-1}\left[\frac{-1}{s^2+1}\right] = \frac{1}{t} L^{-1}\left[\frac{1}{1+s^2}\right] = \frac{1}{t}\sin t$$

Example .7 **Find** $L^{-1}\left[\log\left(\frac{s^2-1}{s^2}\right)\right]$

Solution : W.K.T. $L^{-1}[\varphi'(s)] = -t\,L^{-1}[\varphi(s)]$

(or) $L^{-1}[\varphi(s)] = -\frac{1}{t}L^{-1}[\varphi'(s)]$

$$L^{-1}\left[\log\left(\frac{s^2-1}{s^2}\right)\right] = -\frac{1}{t}L^{-1}\left[\frac{d}{ds}\left(\log\frac{s^2-1}{s^2}\right)\right]$$

$$= -\frac{1}{t}L^{-1}\left[\frac{d}{ds}[\log(s^2-1)-\log s^2]\right]$$

$$= -\frac{1}{t}L^{-1}\left[\frac{2s}{s^2-1}-\frac{2s}{s^2}\right] = -\frac{1}{t}L^{-1}\left[\frac{2s}{s^2-1}-\frac{2}{s}\right]$$

$$= -\frac{2}{t}L^{-1}\left[\frac{s}{s^2-1}-\frac{1}{s}\right] = -\frac{2}{t}[\cosh t-1]$$

$$= \frac{2}{t}[1-\cosh t]$$

Example 8 **Find** $L^{-1}\left[\log\left(\frac{s+1}{s-1}\right)\right]$

Solution : W.K.T. $L^{-1}[\varphi(s)] = -\frac{1}{t}L^{-1}[\varphi'(s)]$

$$L^{-1}\left[\log\left(\frac{s+1}{s-1}\right)\right] = -\frac{1}{t}L^{-1}\left[\frac{d}{ds}\left[\log\left(\frac{s+1}{s-1}\right)\right]\right]$$

$$= -\frac{1}{t}L^{-1}\left[\frac{d}{ds}[\log(s+1)-\log(s-1)]\right]$$

$$= -\frac{1}{t}L^{-1}\left[\frac{1}{s+1}-\frac{1}{s-1}\right] = -\frac{1}{t}[e^{-t}-e^{t}]$$

$$= \frac{1}{t}\left[e^{t}-e^{-t}\right] = \frac{2}{t}\left[\frac{e^{t}-e^{-t}}{2}\right] = \frac{2}{t}\sinh t$$

♦ INVERSE LAPLACE TRANSFORM OF INTEGRALS.

$$L^{-1}\left[\int_s^\infty \varphi(s)\,ds\right] = \frac{1}{t}f(t) = \frac{1}{t}L^{-1}[\varphi(s)]$$

(or) $$L^{-1}[\varphi(s)] = t\,L^{-1}\left[\int_s^\infty \varphi(s)\,ds\right]$$

PROBLEMS BASED ON INVERSE LAPLACE TRANSFORM OF INTEGRALS

Example 9 Obtain $L^{-1}\left[\dfrac{2s}{(s^2+1)^2}\right]$

Solution : W.K.T. $L^{-1}[\varphi(s)] = tL^{-1}\left[\int_s^{\infty}\varphi(s)\,ds\right]$

$$L^{-1}\left[\frac{2s}{(s^2+1)^2}\right] = t\,L^{-1}\left[\int_s^{\infty}\frac{2s}{(s^2+1)^2}\,ds\right]$$

$$= t\,L^{-1}\left[\left(\frac{-1}{s^2+1}\right)_s^{\infty}\right] = t\,L^{-1}\left[0+\frac{1}{s^2+1}\right]$$

$$= t\,L^{-1}\left[\frac{1}{s^2+1}\right] = t\sin t$$

PROBLEMS BASED ON PARTIAL FRACTION

Example 9 Find $L^{-1}\left[\dfrac{1-s}{(s+1)(s^2+4s+13)}\right]$

Solution : $\dfrac{1-s}{(s+1)(s^2+4s+13)} = \dfrac{A}{s+1}+\dfrac{Bs+C}{s^2+4s+13}$

$1-s = A(s^2+4s+13)+(Bs+c)+(s+1)$

Put $s = -1$ we get $1+1 = A(1-4+13)$

$2 = 10A$

$A = 1/5$

Put $s = 0$ we get $1 = 13\,A + C$

$1 = \dfrac{13}{5} + C$

$C = 1-\dfrac{13}{5}$ $\quad\therefore C = -\dfrac{8}{5}$

Equating the coefficients of s^2 on both sides

$0 = A + B$

$B = -A$

$B = -\dfrac{1}{5}$

$$\frac{1-s}{(s+1)(s^2+4s+13)} = \frac{1/5}{s+1} + \frac{-1/5\,s-8/5}{s^2+4s+13}$$

$$= \frac{1}{5}\left[\frac{1}{s+1}\right] - \frac{1}{5}\left[\frac{s+8}{s^2+4s+13}\right]$$

$$= \frac{1}{5}\left[\frac{1}{s+1}\right] - \frac{1}{5}\left[\frac{s+8}{(s+2)^2+13-4}\right]$$

$$= \frac{1}{5}\left[\frac{1}{s+1}\right] - \frac{1}{5}\left[\frac{s+2}{(s+2)^2+3^2} + \frac{6}{(s+2)^2+3^2}\right]$$

$$L^{-1}\left[\frac{1-s}{(s+1)(s^2+4s+13)}\right]$$

$$= \frac{1}{5}L^{-1}\left[\frac{1}{s+1}\right] - \frac{1}{5}L^{-1}\left[\frac{s+2}{(s+2)^2+3^2}\right] - \frac{2}{5}L^{-1}\left[\frac{3}{(s+2)^2+3^2}\right]$$

$$= \frac{1}{5}e^{-t} - \frac{1}{5}e^{-2t}L^{-1}\left[\frac{s}{s^2+3^2}\right] - \frac{2}{5}e^{-2t}L^{-1}\left[\frac{3}{s^2+3^2}\right]$$

$$= \frac{1}{5}e^{-t} - \frac{1}{5}e^{-2t}\cos 3t - \frac{2}{5}e^{-2t}\sin 3t$$

Example 10 Find $L^{-1}\left[\frac{5s^2-15s-11}{(s+1)(s-2)^3}\right]$

Solution : $\frac{5s^2-15s-11}{(s+1)(s-2)^3} = \frac{A}{s+1} + \frac{B}{s-2} + \frac{C}{(s-2)^2} + \frac{D}{(s-2)^3}$

$5s^2-15s-11 = A(s-2)^3+B(s+1)(s-2)^2+C(s+1)(s-2)+D(s+1)$

Put $s = -1$ we get

$$5+15-11 = A(-1-2)^3$$
$$9 = -27A$$
$$A = -\frac{1}{3}$$

equating the coefficients of s^3 on both sides

$$0 = A + B$$
$$B = -A$$
$$B = \frac{1}{3}$$

Put $s = 2$ we get

$$-21 = 3D$$
$$D = -7$$

Put $s = 0$ we get

$$-11 = -8A + 4B - 2C + D$$
$$= -8\left(-\frac{1}{3}\right) + 4\left(\frac{1}{3}\right) - 2C - 7$$
$$-4 = \frac{8}{3} + \frac{4}{3} - 2C$$
$$= 4 - 2C$$
$$-8 = -2C$$
$$C = 4$$

$$\frac{5s^2 - 15s - 11}{(s+1)(s-2)^3} = \frac{-\frac{1}{3}}{s+1} + \frac{\frac{1}{3}}{s-2} + \frac{4}{(s-2)^2} - \frac{7}{(s-2)^3}$$

$$L^{-1}\left[\frac{5s^2 - 15s - 11}{(s+1)(s-2)^3}\right] = -\frac{1}{3}L^{-1}\left[\frac{1}{s+1}\right] + \frac{1}{3}L^{-1}\left[\frac{1}{s-2}\right] + 4L^{-1}\left[\frac{1}{(s-2)^2}\right] - 7L^{-1}\left[\frac{1}{(s-2)^3}\right]$$

$$= -\frac{1}{3}e^{-t} + \frac{1}{3}e^{2t} + 4e^{2t}L^{-1}\left[\frac{1}{s^2}\right] - 7e^{2t}L^{-1}\left[\frac{1}{s^3}\right]$$

♦ SECOND SHIFTING PROPERTY

$$L^{-1}\left[e^{-as}F(s)\right] = f(t-a)\,U(t-a)$$

PROBLEMS BASED ON INVERSE LAPLACE TRANSFORM [SECOND SHIFTING PROPERTY]

Example 5.7.54. Find $L^{-1}\left[\frac{e^{-\pi s}}{s+3}\right]$

Solution : $L^{-1}\left[\frac{1}{s+3}\right] = e^{-3t}$

$$L^{-1}\left[\frac{e^{-\pi s}}{s+3}\right] = e^{-3(t-\pi)}\,U(t-\pi)$$

Example 12 Find $L^{-1}\left[\frac{1}{\sqrt{1+s^2}}\right]$

Solution : $$\frac{1}{\sqrt{1+s^2}} = \frac{1}{\sqrt{s^2\left[1+\frac{1}{s^2}\right]}}$$

$$= \frac{1}{s\sqrt{1+\frac{1}{s^2}}} = \frac{1}{s}\left[1+\frac{1}{s^2}\right]^{-1/2}$$

$$= \frac{1}{s}\left[1-\frac{1}{2}\frac{1}{s^2}+\frac{1.3}{2^2}\frac{1}{2!}\frac{1}{s^4}-\frac{1.3.5}{2^3}\frac{1}{3!}\frac{1}{s^6}+\dots\right]$$

$$= \frac{1}{s}-\frac{1}{2}\frac{1}{s^3}+\frac{1.3}{2^2}\frac{1}{2!}\frac{1}{s^5}-\frac{1.3.5}{2^3}\frac{1}{3!}\frac{1}{s^7}+\dots$$

$$L^{-1}\left[\frac{1}{\sqrt{1+s^2}}\right] = L^{-1}\left[\frac{1}{s}\right]-\frac{1}{2}L^{-1}\left[\frac{1}{s^3}\right]+\frac{1.3}{2^2}\frac{1}{2!}L^{-1}\left[\frac{1}{s^5}\right]-\dots$$

$$= 1-\frac{1}{2}\frac{1}{2!}t^2+\frac{1.3}{2^2}\frac{1}{2!}\frac{1}{4!}t^4-\frac{1.3.5}{2^3}\frac{1}{3!}\frac{1}{6!}t^6+\dots$$

$$= 1-\frac{t^2}{4}+\frac{t^4}{64}-\frac{t^6}{384}+\dots$$

♦ **CHANGE OF SCALE PROPERTY.**

STATE AND PROVE THE SCALLING PROPERTY OF L.T.

PROBLEMS BASED ON INVERSE LAPLACE TRANSFORM
CHANGE OF SCALE PROPERTY

Example 13 If $L f(t) = F(s)$ then $L[f(at)] = \frac{1}{a} F\left[\frac{s}{a}\right]$

If $f(t) = L^{-1}[F(s)]$ then $L^{-1}[F(cs)] = \frac{1}{c} f\left(\frac{t}{c}\right)$

Solution : $$L[f(t)] = \int_0^\infty e^{-st} f(t)\, dt$$

$$L[f(at)] = \int_0^\infty e^{-st} f(at)\, dt$$

Put $at = x \qquad t \to 0 \qquad \Rightarrow x \to 0$

$adt = dx \qquad t \to \infty \qquad \Rightarrow x \to \infty$

$$= \int_0^\infty e^{-s x/a} f(x) \frac{dx}{a} = \frac{1}{a} \int_0^\infty e^{-(s/a)x} f(x)\, dx$$

$$= \frac{1}{a} \int_0^\infty e^{-(s/a)t} f(t)\, dt \qquad [\because x \text{ is dummy variable}]$$

$$= \frac{1}{a} F\left[\frac{s}{a}\right]$$

$$L^{-1}\left[\frac{1}{a} F\left(\frac{s}{a}\right)\right] = f(at)$$

$$\frac{1}{a} L^{-1}\left[F\left(\frac{s}{a}\right)\right] = f(at)$$

$$L^{-1}\left[F\left(\frac{s}{a}\right)\right] = a f(at)$$

Put $a = \frac{1}{c}$ we get $L^{-1}[F(cs)] = \frac{1}{c} f\left(\frac{t}{c}\right)$

Example 14 If $L[f(t)] = F(s)$ find $L\left[f\left(\frac{t}{a}\right)\right]$

Solution : $$L[f(t)] = \int_0^\infty e^{-st} f(t)\, dt$$

$$L[f(t/a)] = \int_0^\infty e^{-st} f\left(\frac{t}{a}\right) dt$$

Put $u = \frac{t}{a} \qquad t \to 0 \Rightarrow u \to 0$

$du = \frac{dt}{a} \qquad t \to \infty \Rightarrow u \to \infty$

$$L[f(t/a)] = \int_0^\infty e^{-s(au)} f(u)\, adu$$

$$= a \int_0^\infty e^{-asu} f(u)\, du$$

$$= a \int_0^{\infty} e^{-ast} f(t)\, dt \qquad [\because u \text{ is a dummy variable}]$$

$$= a\ F[as]$$

CONVOLUTION THEOREM

Define convolution.

Solution .

The convolution of two functions $f(t)$ and $g(t)$ is defined as

$$f(t) * g(t) = \int_0^t f(u)\, g(t-u)\, du$$

Example 14 . Prove that f (t) * g (t) = g (t) * f (t).

Solution : $f(t) * g(t) = \int_0^t f(u)\, g(t-u)\, du$

$$\text{W.K.T. } \int_0^a f(x)\, dx = \int_0^a f(a-x)\, dx$$

$$= \int_0^t f(t-u)\, g\,[t-(t-u)]\, du = \int_0^t f(t-u)\, g(u)\, du$$

$$= \int_0^t g(u) f(t-u)\, du = g(t) * f(t)$$

State and prove convolution theorem.

Solution : If $f(t)$ and $g(t)$ are functions defined for $t \geq 0$

then $L\,[f(t) * g(t)] = L\,[f(t)]\,.\,L\,[g(t)]$

Proof : We know that $L\,[f(t)] = \int_0^{\infty} e^{-st} f(t)\, dt$

$$L\,[f(t) * g(t)] = \int_0^{\infty} e^{-st}\, [f(t) * g(t)]\, dt$$

$$= \int_0^{\infty} e^{-st}\, [\int_0^t f(u)\, g(t-u)\, du\,]\, dt$$

by def. of convolution

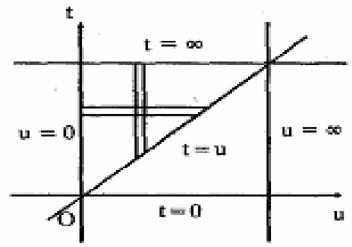
$$= \int_0^{\infty} \int_0^{t} e^{-st} f(u)\, g(t-u)\, du\, dt$$

Change the order of the integration

Given $\quad t = 0$ to $t = \infty$

$\qquad\quad u = 0$ to $u = t$

$$= \int_0^{\infty} \int_u^{\infty} e^{-st} f(u)\, g(t-u)\, dt\, du$$



$$= \int_0^{\infty} f(u) \int_u^{\infty} e^{-st} g(t-u)\, dt\, du$$

put $t - u = v$, $dt = dv$ $\quad\Big|\quad t \to u \Rightarrow v \to 0$, $\quad t \to \infty \Rightarrow v \to \infty$

$$= \int_0^{\infty} f(u) \int_0^{\infty} e^{-(u+v)s} g(v)\, dv\, du$$

$$= \int_0^{\infty} f(u)\, e^{-us} \int_0^{\infty} e^{-vs} g(v)\, dv\, du$$

$$= \int_0^{\infty} f(u)\, e^{-us} du \int_0^{\infty} e^{-vs} g(v)\, dv \quad [\because u \text{ and } v \text{ are dummy variable}]$$

$$= \int_0^{\infty} e^{-st} f(t)\, dt \int_0^{\infty} e^{-st} g(t)\, dt$$

$$= L[f(t) * g(t)] = L[f(t)] \cdot L[(g(t)] = F(s) \cdot G(s)$$

where $\quad L[f(t)] = F(s)$

$\qquad\quad L[g(t)] = G(s)$

$$L^{-1}[F(s)\ G(s)] = f(t) * g(t)$$

$$= L^{-1}[F(s)] * L^{-1}[G(s)]$$

PROBLEMS BASED ON CONVOLUTION THEOREM

Example 15 **Using convolution theorem find** $L^{-1}\left[\dfrac{1}{(s+a)(s+b)}\right]$

Solution :

$$L^{-1}[F(s).\ G(s)] = L^{-1}[F(s)] * L^{-1}[G(s)]$$

$$L^{-1}\left[\frac{1}{s+a}\cdot\frac{1}{s+b}\right] = L^{-1}\left[\frac{1}{s+a}\right] * L^{-1}\left[\frac{1}{s+b}\right]$$

$$= e^{-at} * e^{-bt}$$

$$f(t) * g(t) = \int_0^{t} f(u)\, g(t-u)\, du$$

$$= \int_0^{t} e^{-au} e^{-b(t-u)} du = \int_0^{t} e^{-au} e^{-bt} e^{bu} du$$

$$= e^{-bt} \int_0^t e^{-(a-b)u} du = e^{-bt} \left[\frac{e^{-(a-b)u}}{-(a-b)} \right]_0^t$$

$$= e^{-bt} \left[\frac{e^{-(a-b)t}}{-(a-b)} - \frac{1}{-(a-b)} \right] = \frac{e^{-bt}}{a-b} [1 - e^{-at} e^{bt}]$$

$$= \frac{1}{a-b} [e^{-bt} - e^{-at}]$$

Example 16 Using convolution theorem find $L^{-1} \left[\frac{s}{(s^2+a^2)^2} \right]$

Solution :

$$L^{-1} [F(s)\ G(s)] = L^{-1} [F(s)] * L^{-1} [G(s)]$$

$$L^{-1} \left[\frac{s}{(s^2+a^2)^2} \right] = L^{-1} \left[\frac{s}{s^2+a^2} \right] * L^{-1} \left[\frac{1}{s^2+a^2} \right]$$

$$= L^{-1} \left[\frac{s}{s^2+a^2} \right] * \frac{1}{a} L^{-1} \left[\frac{a}{s^2+a^2} \right]$$

$$= \cos at * \frac{1}{a} \sin at = \frac{1}{a} [\cos at * \sin at]$$

$$= \frac{1}{a} \int_0^t \cos au \ \sin a(t-u)\, du$$

$$= \frac{1}{a} \int_0^t \sin(at - au) \ \cos au\, du$$

$$= \frac{1}{a} \int_0^t \frac{\sin(at - au + au) + \sin(at - au - au)}{2} du$$

$$= \frac{1}{2a} \int_0^t [\sin at + \sin a(t - 2u)]\, du$$

$$= \frac{1}{2a} \left[(\sin at)\, u + \left(\frac{-\cos a(t-2u)}{-2a} \right) \right]_0^t$$

$$= \frac{1}{2a} \left[u (\sin at) + \frac{\cos a(t-2u)}{2a} \right]_0^t$$

$$= \frac{1}{2a} \left[\left(t \sin at + \left(\frac{\cos at}{2a} \right) - \left(0 + \frac{\cos at}{2a} \right) \right) \right]$$

$$= \frac{1}{2a} \left[t \sin at + \frac{\cos at}{2a} - \frac{\cos at}{2a} \right] = \frac{1}{2a}\, t \sin at$$

Example 17 Find $L^{-1}\left[\frac{s^2}{(s^2+a^2)(s^2+b^2)}\right]$ using convolution theorem.

Solution :

$$L^{-1}[F(s)\,.\,G(s)] = L^{-1}[F(s)] * L^{-1}[G(s)]$$

$$L^{-1}\left[\frac{s}{s^2+a^2}\cdot\frac{s}{s^2+b^2}\right] = L^{-1}\left[\frac{s}{s^2+a^2}\right] * L^{-1}\left[\frac{s}{s^2+b^2}\right]$$

$$= \cos at * \cos bt$$

$$= \int_0^t \cos au \cos b(t-u)\,du$$

$$= \frac{1}{2}\int_0^t [\cos(au+bt-bu)+\cos(au-bt+bu)\,du$$

$$= \frac{1}{2}\int_0^t \left[\cos[(a-b)u+bt]+\cos[(a+b)u-bt]\right] du$$

$$= \frac{1}{2}\left[\frac{\sin(bt+(a-b)u)}{a-b}+\frac{\sin[(a+b)u-bt]}{a+b}\right]_0^t$$

$$= \frac{1}{2}\left[\left(\frac{\sin(bt+at-bt)}{a-b}+\frac{\sin(at+bt-bt)}{a+b}\right) - \left(\frac{\sin bt}{a-b}-\frac{\sin bt}{a+b}\right)\right]$$

$$= \frac{1}{2}\left[\frac{\sin at}{a-b}+\frac{\sin at}{a+b}-\frac{\sin bt}{a-b}+\frac{\sin bt}{a+b}\right]$$

$$= \frac{1}{2}\left[\frac{2a\sin at}{a^2-b^2}-\frac{2b\sin bt}{a^2-b^2}\right] = \frac{1}{2}\left[\frac{2a\sin at-2b\sin bt}{a^2-b^2}\right]$$

$$= \frac{a\sin at-b\sin bt}{a^2-b^2}$$

♦ Solving of Integral equations of convolution type.

Definition : An integral equation of the form

$$y(t) = f(t) + \int_0^t F(t-u)\, G(u)\, du$$

is called integral equation of convolution type.

This equation can also be expressed as

$$y(t) = f(t) + F(t) * G(t).$$

PROBLEMS BASED ON INTEGRAL EQUATIONS OF CONVOLUTION TYPE

Example 17 **Solve the integral eqn.**

$$\mathbf{y(t) = 1 + \int_0 y(u) \sin(t-u)\, du}$$

Solution : The given eqn. can be written as

$$y(t) = 1 + y(t) * \sin t$$

$$L[y(t)] = L[1] + L[y(t) * \sin t]$$

$$= \frac{1}{s} + L[y(t)]\, L[\sin t]$$

$$= \frac{1}{s} + L[y(t)] \left[\frac{1}{s^2+1}\right]$$

$$\left[1 - \frac{1}{s^2+1}\right] L[y(t)] = \frac{1}{s}$$

$$\left[\frac{s^2}{s^2+1}\right] L[y(t)] = \frac{1}{s}$$

$$L[y(t)] = \frac{s^2+1}{s^3} = \frac{1}{s} + \frac{1}{s^3}$$

$$y(t) = L^{-1}\left[\frac{1}{s}\right] + L^{-1}\left[\frac{1}{s^3}\right] = 1 + \frac{1}{2}t^2$$

SOLUTION OF LINEAR ODE OF SECOND ORDER WITH CONSTANT COEFFICIENTS AND FIRST ORDER SIMULTANEOUS EQUATIONS WITH CONSTANT COEFFICIENTS USING LAPLACE TRANSFORMATION.

PROBLEMS BASED ON SOLUTION OF LINEAR ODE OF SECOND ORDER WITH CONSTANT COEFFICIENTS

Example 18 Solve by using L.T. $(D^2+9)y = \cos 2t$, given that if $y(0) = 1$, $y\left(\frac{\pi}{2}\right) = -1$

Solution : Given $(D^2+9)\,y = \cos 2t$ i.e., $y''(t) + 9y(t) = \cos 2t$

Taking Laplace transforms on both sides,

$$L[y''(t)] + 9L[y(t)] = L[\cos 2t]$$

$$s^2 L[y(t)] - sy(0) - y'(0) + 9L[y(t)] = \frac{s}{s^2+4}$$

Using the initial conditions

$y(0) = 1$, and taking $y'(0) = k$

we have

$$s^2 L[y(t)] - (s)(1) - k + 9L[y(t)] = \frac{s}{s^2+4}$$

$$(s^2+9)L[y(t)] = \frac{s}{s^2+4} + s + k$$

$$L[y(t)] = \frac{s}{(s^2+4)(s^2+9)} + \frac{s+k}{s^2+9} \quad \text{... (1)}$$

$$\frac{s}{(s^2+4)(s^2+9)} = \frac{As+B}{s^2+4} + \frac{Cs+D}{s^2+9} \quad \text{... (A)}$$

$$s = (As+B)(s^2+9) + (Cs+D)(s^2+4)$$

equating s^3 on bothsides we get $0 = A + C$ ... (2)

equating s^2 on bothsides we get $0 = B + D$ ... (3)

Put $s = 0$ on bothsides we get $0 = 9B + 4D$ ... (4)

equating s on bothsides we get $1 = 9A + 4C$ ... (5)

$$(2) \Rightarrow C = -A$$

$$\therefore \quad (5) \Rightarrow 9A + 4(-A) = 1$$

$$9A - 4A = 1$$

$$5A = 1$$

$$A = \frac{1}{5}, C = -\frac{1}{5}$$

$$(3) \Rightarrow \qquad D = -B$$

$$(4) \Rightarrow 9B + 4(-B) = 0$$

$$9B - 4B = 0$$

$$5B = 0$$

$$B = 0, D = 0$$

$$(A) \Rightarrow \frac{s}{(s^2+4)(s^2+9)} = \frac{\frac{1}{5}s + 0}{s^2+4} + \frac{-\frac{1}{5}s + 0}{s^2+9}$$

$$= \frac{s}{5(s^2+4)} - \frac{s}{5(s^2+9)}$$

$$\therefore (1) \Rightarrow L[y(t)] = \frac{s}{5(s^2+4)} - \frac{s}{5(s^2+9)} + \frac{s}{s^2+9} + \frac{k}{s^2+9}$$

$$y(t) = \frac{1}{5}L^{-1}\left[\frac{s}{s^2+4}\right] - \frac{1}{5}L^{-1}\left[\frac{s}{s^2+9}\right]$$

$$+ L^{-1}\left[\frac{s}{s^2+9}\right] + kL^{-1}\left[\frac{1}{s^2+9}\right]$$

$$= \frac{1}{5}\cos 2t - \frac{1}{5}\cos 3t + \cos 3t + \frac{k}{3}\sin 3t$$

Put $t = \frac{\pi}{2}$ we get $y\left(\frac{\pi}{2}\right) = \frac{1}{5}(-1) - \frac{1}{5}(0) + 0 + \frac{k}{3}(-1) = -\frac{1}{5} - \frac{k}{3}$

But given $y\left(\frac{\pi}{2}\right) = -1 \qquad \therefore -1 = -\frac{1}{5} - \frac{k}{3}$

$$1 = \frac{1}{5} + \frac{k}{3}$$

$$1 - \frac{1}{5} = \frac{k}{3}$$

$$\frac{4}{5} = \frac{k}{3} \Rightarrow k = \frac{12}{5}$$

$$\therefore \quad y(t) = \cos 3t + \frac{4}{5}\sin 3t + \frac{1}{5}\cos 2t - \frac{1}{5}\cos 3t$$

$$y(t) = \frac{4}{5}[\cos 3t + \sin 3t] + \frac{1}{5}\cos 2t$$

Example 19 Solve $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0$ given that $y = \frac{dy}{dx} = 1$ at $x = 0$ using L.T. method.

Solution : Given $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0$ and $y(0) = 1,\ y'(0) = 1$

i.e., $y''(x) - 2y'(x) + 2y(x) = 0$

$$L[y''(x)] - 2L(y'(x)] + 2L[y(x)] = 0$$

$$s^2 L[y(x) - s\,y(0) - y'(0)] - 2[s\,L[y(x)] - y(0)] + 2L[y(x)] = 0$$

$$s^2 L[y(x) - s - 1] - 2[s\,L[y(x)] - 1] + 2L[y(x)] = 0$$

$$[s^2 - 2s + 2]\,L[y(x)] - s - 1 + 2 = 0$$

$$[s^2 - 2s + 2]\,L[y(x)] = s - 1$$

$$L[y(x)] = \frac{s-1}{s^2 - 2s + 2} = \frac{s-1}{(s-1)^2 + 1}$$

$$y(x) = L^{-1}\left[\frac{s-1}{(s-1)^2+1}\right] = e^t L^{-1}\left[\frac{s}{s^2+1}\right] = e^t \cos t$$

Example 20 Using L.T solve $y'' - 3y' + 2y = e^{-t}$ given $y(0) = 1$, $y'(0) = 0$.

Solution : $y'' - 3y' + 2y = e^{-t}$ and $y(0) = 1,\ y'(0) = 0$

$$L[y''(t)] - 3L[y'(t)] + 2L[y(t)] = L[e^{-t}]$$

$$s^2 L[y(t)] - s\,y(0) - y'(0) - 3[s\,L[y(t)] - y(0)] + 2L[y(t)] = \frac{1}{s+1}$$

$$s^2 L[y(t)] - s - 0 - 3s\,L[y(t)] + 3 + 2L[y(t)] = \frac{1}{s+1}$$

$$(s^2 - 3s + 2)\,L[y(t)] = \frac{1}{s+1} + s - 3$$

$$(s-1)(s-2)\,L[y(t)] = \frac{s^2-2s-2}{s+1}$$

$$L[y(t)] = \frac{s^2-2s-2}{(s+1)(s-1)(s-2)} = \frac{A}{s+1} + \frac{B}{s-1} + \frac{C}{s-2}$$

$$s^2-2s-2 = A(s-1)(s-2) + B(s+1)(s-2) + C(s+1)(s-1)$$

Put $s = 1$ we get $1-2-2 = -2B$

$$-3 = -2B$$

$$B = 3/2$$

Put $s = 2$ we get $4-4-2 = 3C$

$$C = -2/3$$

Put $s = 1$ we get $1+2-2 = 6A$

$$A = 1/6$$

[redacted]

Example 21 Using L.T. Solve $\frac{dx}{dt} + 3x - 2y = 1$; $\frac{dy}{dt} - 2x + 3y = e^t$ given that $x = 0 = y$ when $t = 0$.

Solution : The given differential equations can be written as

$$x'(t) + 3x(t) - 2y(t) = 1$$

$$y'(t) - 2x(t) + 3y(t) = e^t$$

Taking L.T. on both sides

$$L[x'(t)] + 3\,L[x(t)] - 2\,L[y(t)] = L[1]$$

$$L[y'(t)] - 2\,L[x(t)] + 3\,L[y(t)] = L[e^t]$$

$$s\,L[x(t)] - x(0) + 3\,L[x(t)] - 2\,L[y(t)] = \frac{1}{s}$$

$$s\,L[y(t)] - y(0) - 2\,L[x(t)] + 3\,L[y(t)] = \frac{1}{s-1}$$

Given $x(0) = 0$, $y(0) = 0$

$$s\,L[x(t)] + 3\,L[x(t)] - 2\,L[y(t)] = \frac{1}{s} \qquad \ldots (1)$$

$$s\,L[y(t)] - 2\,L[x(t)] + 3\,L[y(t)] = \frac{1}{s-1} \qquad \ldots (2)$$

$$(1) \Rightarrow \quad (s+3)\,L[x(t)] - 2L[y(t)] = \frac{1}{s} \qquad \dots (3)$$

$$-2\,L[x(t)] + (s+3)\,L[y(t)] = \frac{1}{s-1} \qquad \dots (4)$$

Solving (3) & (4) we get

$$L[x(t)] = \frac{\begin{vmatrix} 1/s & -2 \\ 1/(s-1) & s+3 \end{vmatrix}}{\begin{vmatrix} s+3 & -2 \\ -2 & s+3 \end{vmatrix}} = \frac{s^2+4s-3}{s(s-1)(s+1)(s+5)}$$

$$\frac{s^2+4s-3}{s(s-1)(s+1)(s+5)} = \frac{A}{s} + \frac{B}{s-1} + \frac{C}{s+1} + \frac{D}{s+5}$$

$$s^2+4s-3 = A(s-1)(s+1)(s+5) + B(s)(s+1)(s+5) + C(s)(s-1)(s+5) + D(s)(s-1)(s+1)$$

Put $s = 0$ we get $-3 = A(-1)(1)(5)$

$$-3 = -5A$$

$$A = \frac{3}{5}$$

Put $s = 1$ we get

$$1+4-3 = 0 + B(1)(2)(6)$$

$$2 = 12\,B$$

$$B = 1/6$$

Put $s = -1$ we get

$$1-4+3 = 0+0+C(-1)(-2)(4)+0$$

$$-6 = 8C$$

$$C = -\frac{3}{4}$$

Put $s = -5$ we get

$$25-20-3 = 0 + 0 + 0 + D(-5)(-6)(-4)$$

$$2 = -120D$$

$$D = -\frac{1}{60}$$

$$L[x(t)] = \frac{s^2+4s-3}{s(s-1)(s+1)(s+5)} = \frac{3/5}{s} + \frac{1/6}{s-1} + \frac{-3/4}{s+1} + \frac{-1/60}{s+5}$$

$$x(t) = \frac{3}{5}L^{-1}\left[\frac{1}{s}\right] + \frac{1}{6}L^{-1}\left[\frac{1}{s-1}\right] - \frac{3}{4}L^{-1}\left[\frac{1}{s+1}\right] - \frac{1}{60}L^{-1}\left[\frac{1}{s+5}\right]$$

$$= \frac{3}{5}(1) + \frac{1}{6}e^{t} - \frac{3}{4}e^{-t} - \frac{1}{60}e^{-5t}$$

Example 22 Solve $\dot{x} + y = \sin t$; $x + \dot{y} = \cos t$ with $x = 2$ and $y = 0$, when t = ..

Solution : Given $x'(t) + y(t) = \sin t$ and $x(0) = 2,\ y(0) = 0$

$$x(t) + y'(t) = \cos t$$

$$L[x'(t)] + L[y(t)] = L[\sin t]$$

$$L[x(t)] + L[y'(t)] = L[\cos t]$$

$$s\ L\ [x(t)] - x(0) + L[y(t)] = \frac{1}{s^2+1}$$

$$L[x(t)] + s\,L[y(t)] - y(0)] = \frac{s}{s^2+1}$$

$$s\ L\ [x(t)] + L[y(t)] = 2 + \frac{1}{s^2+1} \qquad \text{... (1)}$$

$$L\ [x(t)] + s\,L[y(t)] = \frac{s}{s^2+1} \qquad \text{... (2)}$$

Solving (1) and (2) we get

$$(1-s^2)\,L[y(t)] = 2 + \frac{1-s^2}{s^2+1}$$

$$(1-s^2)\,L[y(t)] = \frac{2s^2+2+1-s^2}{s^2+1}$$

$$L[y(t)] = \frac{s^2+3}{(s^2+1)(1-s^2)}$$

$$\frac{s^2+3}{(s^2+1)(1-s^2)} = \frac{As+B}{s^2+1}+\frac{Cs+D}{1-s^2}$$

$$s^2+3 = (As+B)(1-s^2)+(Cs+D)(s^2+1)$$

Equating s^3 on bothsides

$$0 = -A+C$$
$$A = C$$

Equating s^2 on bothsides

$$1 = -B+D$$

Equating s on both sides

$$0 = A+C$$

Put $s = 0$

$$3 = B+D$$
$$\therefore\ A = 0$$
$$C = 0$$
$$D = 2$$
$$B = 1$$

$$\Rightarrow y(t) = L^{-1}\left[\frac{1}{s^2+1}\right]-2L^{-1}\left[\frac{1}{s^2-1}\right]$$
$$= \sin t - 2\sinh t$$

To find $x(t)$, we have

$$x(t)+y'(t) = \cos t$$
$$x(t) = \cos t - y'(t)$$
$$y(t) = \sin t - 2\sinh t$$
$$\frac{dy}{dt} = \cos t - 2\cosh t$$
$$\therefore\ x(t) = \cos t - \cos t + 2\cosh t$$
$$= 2\cosh t$$

Hence $x(t) = 2\cosh t,\ y(t) = \sin t - 2\sinh t$